

Shabd Srivastava

+91 8736914818 Shabdsrivastava19@gmail.com

[LinkedIn](#) [GitHub](#) [Portfolio](#)

Work Experience

Rebeca, Annual Fest — IEST Shibpur

Web Development Associate

On-campus

January 2026 – April 2026

- Built the fest website end-to-end with a responsive UI — held up fine through 500+ people on it at once during peak registration.
- Set up registration and submission forms for 10+ events so the core team didn't have to collect everything by hand.
- Handled deployment and hosting myself, including fixing things live during the busiest days of the fest.
- Worked with the branding team to keep the site in line with the rest of the fest's design and messaging.

Projects

Shawty — URL Shortener (System Design)

[Github](#)

- **Redis-backed caching, rate limiting, and async click logging** — reads fail open to Postgres if Redis is down, and custom short codes are race-safe through a Postgres UNIQUE constraint, not just an app-level check.
- **Link expiration returns a proper 410**, and the TTL is checked on every cache hit so a stale cache entry can't outlive Postgres truth.
- **Runs as 3 replicas behind nginx** in Docker Compose; also wrote Kubernetes manifests (probes, HPA, resource limits) and validated them end-to-end on a local kind cluster.
- **34 unit and 13 integration tests**, CI green via GitHub Actions; the analytics endpoint runs one CTE-based SQL query instead of pulling every row and aggregating in app code.

Stack: TypeScript · Express · PostgreSQL · Redis · Docker · Kubernetes · nginx · GitHub Actions

DocuMind — AI-Powered Document Intelligence Platform

[Live](#) [Github](#)

- **Full-stack app** where users upload PDFs and get back LLM-generated summaries, flashcards, and study notes.
- **Built the document pipeline to fail gracefully** — typed failure states, retry logic, and fallback persistence for when the AI provider errors out or rate-limits.
- **Used Supabase** for auth and Postgres storage with row-level security, with modular REST APIs for extraction and generation.
- **Migrated** the AI provider from Gemini to OpenRouter after tracking down a PDF parsing dependency that silently broke in the serverless environment despite working fine locally.
- **Deployed** on Vercel with environment-based config and CI/CD.

Stack: Next.js · React · TypeScript · Tailwind CSS · Supabase · OpenRouter · PDF Parsing · REST APIs

DiseaseDetect — Explainable Medical Image Classification

[Live](#) [Github](#)

- **Trained and compared** DenseNet121, ResNet50, and EfficientNetB0 for pneumonia detection from chest X-rays — DenseNet121 came out on top at 92% ROC-AUC.
- **Added a Grad-CAM layer** so you can see which parts of the X-ray the model is actually keying off, instead of trusting a black box.
- **Built modular TensorFlow pipelines** with augmentation ablation, reproducible splits, checkpointing, and TensorBoard tracking across all three models.
- **Evaluated** everything on ROC-AUC, F1, precision, recall, confusion matrices, and FP/FN breakdowns, with the tradeoffs between architectures written up.
- **Deployed** as a Streamlit app on Hugging Face Spaces — upload an X-ray, get a prediction with the Grad-CAM overlay.

Stack: Python · TensorFlow · Keras · CNNs · Transfer Learning · Grad-CAM · Streamlit · Hugging Face Spaces

Skills

Languages: Python, Java, SQL, TypeScript, JavaScript, C, C++

Frameworks & Tools: React, Next.js, Node.js, Express, TensorFlow, Keras, Supabase (Postgres), Docker, Kubernetes, Streamlit, Tailwind CSS, Git

Domains: Full-Stack Dev, LLM Integration, Deep Learning, Computer Vision, REST APIs, CI/CD, System Design, Model Deployment

Education

Indian Institute of Engineering Science & Technology, Shibpur

B.Tech, Information Technology

West Bengal, India

2024 – 2028